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B.Sc. Biochemistry &
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BACHELOR THESIS

Physiological and geodata analyses of methane reduction potential in ruminants

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Handed in: [Date]

ABSTRACT

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LIST OF ABBREVIATIONS

6-Aza	6-Azauracil.
8-Aza	8-Azahypoxanthin.
AB	antibiotic.
CFU	colony forming unit.
DMSO	dimethylsulfoxid.
EtOH	ethanol.
<i>M. boviskoreani</i>	<i>Methanobrevibacter boviskoreani</i> AbM4.
<i>M.</i>	<i>Methanobrevibacter</i> .
<i>M. millerae</i>	<i>Methanobrevibacter millerae</i> ZA-10 ^T .
<i>M. thaueri</i>	<i>Methanobrevibacter thaueri</i> CW ^T .
<i>M. wolinii</i>	<i>Methanobrevibacter wolinii</i> SH ^T .
OD600	optical density at 600 nm.
Puro	puromycin.
rRNA	ribosomal RNA.
Sqr.	square.

1 INTRODUCTION

The following section presents the results regarding a potential mitigation strategy against methane emissions: the immunization of cattle ([Baca-González et al., 2020](#)). As explained above, the bachelor's thesis - and thus the results - can be divided into two main sections:

1. The biochemical part, which involves a fundamental analysis of methane-producing archaea isolated from the rumen of cattle, as well as an attempt to develop genetic tools for them.
2. The geographical part, which, based on [Buendia et al. \(2019\)](#) findings, examines how much methane can be reduced and what role this can play in combating climate change.

In the enteric fermentation process of ruminants CH_4 are formed through methanogenic archaea. One discussed mitigation strategy to reduce methane resulting from cattle farming is to immunize the cattle against these specific archaea ([Malyugina et al., 2025](#)). For developing a specific immunization strategy, the isolated methanogenic archaea strains from cattle rumen has to be well understood. Although some strains have already been isolated from the rumen of cattle, they have hardly been studied to date ([Baca-González et al., 2020](#)). This bachelor's thesis aims to address this issue.

In doing so, the focus will primarily be on two strains: *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*). Besides these two main strains, some experiments were also done on *Methanobrevibacter wolinii* SH^T (*M. wolinii*) and *Methanobrevibacter millerae* ZA-10^T (*M. millerae*). These two methanogenic archaea were also isolated from the rumen of cattle. Goal of the investigation, that took place, was to detect several basic characteristics and try to implement first genetic engineering tools.

2 DATA

2.1 *Section name*

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3 METHODS

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Table 2: Example table

LEFT TEXT	CENTER TEXT	RIGHT TEXT
A variable name	0.525*** (0.004)	Right text
A multicolumn is good for presenting panels.		
Another variable	0.323 (0.419)	Right text again

Notes: Table notes should guide a reader to understand the table.
Standard errors in parenthesis, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4 RESULTS

4.1 Biochemistry

All following experiments were done with *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) and *Methanobrevibacter thaueri* CW^T (*M. thaueri*), in some *Methanobrevibacter wolinii* SH^T (*M. wolinii*) and *Methanobrevibacter millerae* ZA-10^T (*M. millerae*) were investigated as well.

4.1.1 Cultivation

Media

For cultivation two liquid, complex media were tested, the DSM 1523 and the DSM 119. Both media are suitable for the cultivation of *M. thaueri* and *M. boviskoreani*, although some specific behavior could be detected.

The *M. thaueri* grew more slowly and not as dense as the *M. boviskoreani*. Also the *M. thaueri* started to lyse rapidly in a few days after full growth. This was observed much more prominently in DSM 1523 than in 119.

M. boviskoreani on the other hand did not show the described lysing behavior but precipitated in both media. These precipitates were not completely dissolvable through shaking. *Here I also could implement the Microscopy picture of the precipitate?*

Substrate

Besides the general cultivation with a H₂/CO₂ headspace, it was tested if *M. thaueri* and *M. boviskoreani* could use ethanol (EtOH) as an alternative source of reducing potential for growth, as a liquid option to hydrogen. For the tested concentration of EtOH (10 mM, 20 mM, 50 mM, 100 mM, 200 mM) no visual growth could be detected. *In einer literatur steht für typstamm bovis er wachse auf EtOH, und für Format auch -> das könnte man reintheoretisch noch testen*

4.1.2 Cell count

Could also show a microscopy picture of cell counting?

Cell counting was done with a phase-contrast microscope and a Neubauer improved counting chamber, only the *M. wolinii* with an optical density at 600 nm (OD₆₀₀) of 0.387 was counted with a Buerker counting chamber. In the Buerker, squares were counted, in the Neubauer

improved, the group squares were counted, both have a volume of $4 \cdot 10^{-6}$ ml, as can be seen in [LO - Laboroptik GmbH \(2026\)](#).

The cell-number per ml for the cultures was then determined using the following calculation:

$$\text{cellnumber} \cdot \text{ml}^{-1} = \frac{\varnothing \text{ cellnumber} \cdot \text{dilution factor}}{\text{Volume of the counted Square}} \quad (1)$$

Table [B1](#) through [B6](#) in the appendix contain the original counted cellnumbers per filled chamber and square. To calculate the cell count per ml, the arithmetic mean was determined for each culture (tab. [B7](#)) and substituted into the equation [1](#) along with the corresponding dilution factor (see *ibid.*). The results of the calculation were recorded in a table [4.1](#). The calculation was performed as an example using *M. wolinii* (OD600 = 0.387):

$$\text{cellnumber} \cdot \text{ml}^{-1} = \frac{216.375 \text{ cells} \cdot 20}{4 \cdot 10^{-6} \text{ ml}} = 1,081,875,000 = 10.819 \cdot 10^8$$

Table 4.1: Cellnumber per ml of different *Methanobrevibacter* (*M.*) culture for corresponding optical density at 600 nm (OD600). For counting the cells a phase-contrast microscope was used and a Buerker or Neubauer improved counting chamber at a 400x magnification. The cell count per ml was calculated using the equation [1](#), and the values were taken from the table [B7](#).

culture	OD600	cellnumber · ml ⁻¹
<i>M. wolinii</i>	0.387	$10.819 \cdot 10^8$
<i>M. wolinii</i>	0.396	$10.672 \cdot 10^8$
<i>M. thaueri</i>	0.235	$5.603 \cdot 10^8$
<i>M. thaueri</i>	0.116	$3.644 \cdot 10^8$
<i>M. millerae</i>	0.378	$8.880 \cdot 10^8$
<i>M. boviskoreani</i>	0.287	$9.272 \cdot 10^8$

A clear trend can be observed for the cellnumber per ml of the different *Methanobrevibacter* (*M.*) culture throughout the table [4.1](#): higher OD600 values correspond to a higher cell count per ml. At the lowest measured OD600 of 0.116 (*M. thaueri*), $3.644 \cdot 10^8$ cells per ml was determined, and at the highest OD600 of 0.396 (*M. wolinii*), $10.672 \cdot 10^8$ cells per ml was determined.

Nevertheless, differences between the organisms are also apparent. For example, *M. boviskoreani* (OD600 = 0.287) has a higher cell count per mL ($9.272 \cdot 10^8$) than *M. millerae* (OD600 = 0.378 and cellnumber per ml = $8.880 \cdot 10^8$) despite having a lower OD600 of nearly 0.1. Furthermore, both *M. wolinii* cultures and *M. millerae* have similar OD600 values between 0.378 and 0.396, but those of *M. wolinii* are around $2 \cdot 10^8$ cells per ml higher (10.819 or $10.672 \cdot 10^8$ to $8.880 \cdot 10^8$ cells per ml).

The *M. wolinii* cultures are quite close to each other in terms of cell count per ml, with the

higher value being determined for the culture with the lower OD600.

Besides *M. wolinii*, two cultures of *M. thaueri* were also examined to determine cell count. Here, one culture has an OD600 of 0.235 that is slightly more than twice as high as the other one with 0.116, but a cell count per ml that is slightly less than twice as high ($5.603 \cdot 10^8$ to $3.644 \cdot 10^8$ cells per ml).

4.1.3 Plating

Spread plating

100 μ l (nochmal im Laborbuch nachschauen) of a full-grown culture were spread plated as duplicates on a DSM 1523 solid media (1.5% agar) under anaerobic condition in a glovebox. In figure 4.1 the plating results are shown.

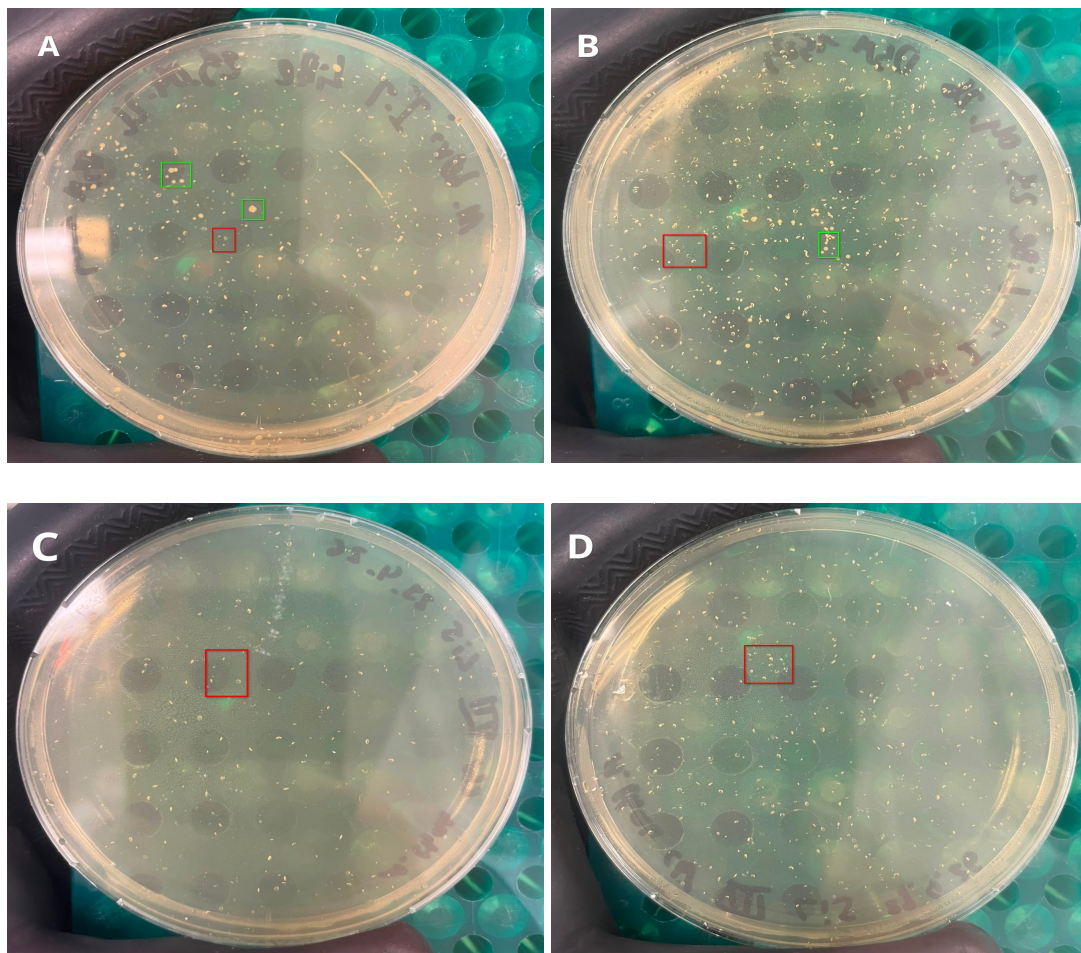


Figure 4.1: Spread plating results of *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*). 100 μ l of a full-grown culture was plated on DSM 1523 (1.5% agar) in duplicates. Red rectangles highlight gas bubble spots, green rectangles grown colonies. A: *M. boviskoreani*, B: *M. boviskoreani*, C: *M. thaueri*, D: *M. thaueri*

Various spots can be seen on all 4 plates. Most of these are gas bubbles, as indicated by the smaller size and transparency of the spots (see red rectangles in Figure 4.1). In contrast,

a few spots are opaque, larger, round, slightly milky, and yellow (see green squares *ibid.*). These are grown colonies. Das die einen Kolonien sind und die anderen Gasblasen ist streng genommen ja bereits eine Interpretation, finde es aber in dem Kontext recht nervig und wenig förderlich für die Verständlichkeit, das nur vage zu bezeichnen als milchig und weniger milchig. Wie siehst du das Max?

No mature colonies of *M. thaueri* were detected on the two plates. *M. boviskoreani*, on the other hand, grew on both plates. Individual colonies were picked from the plates, propagated in liquid DSM 1523, and verified as *M. boviskoreani* via 16S-Sequencing.

Pour plating

The pour plating was not successful for 100 µl of a full-grown *M. thaueri* culture with DSM 119 (1% agar?) as you can see in fig. A1. This experiment was also performed as duplicate.

Roll Plating

200 µl of full-grown *M. thaueri*, *M. wolinii* and *M. boviskoreani* culture have been roll plated in DSM 1523(1% agar?).

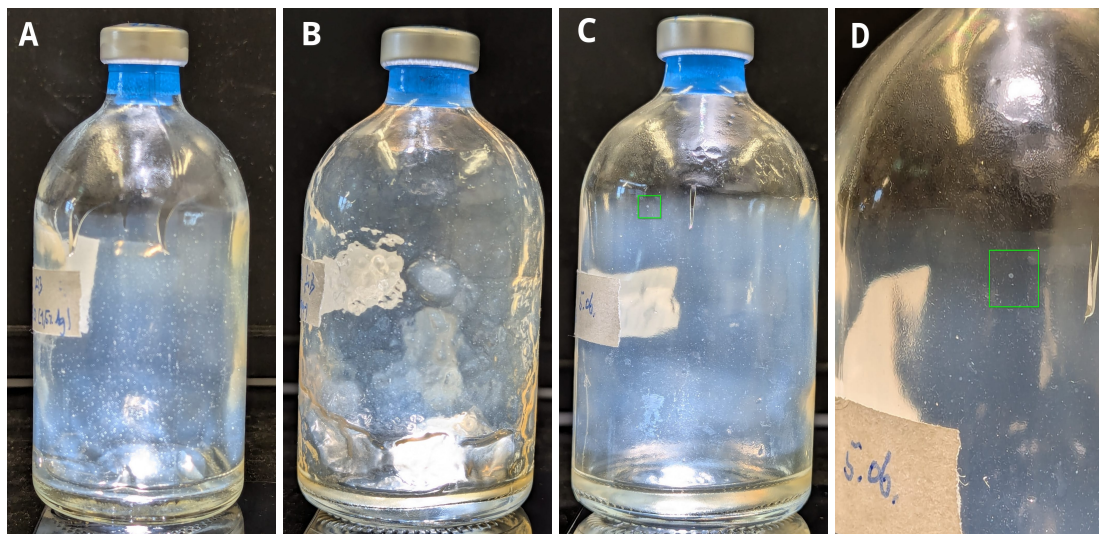


Figure 4.2: Roll plating of *Methanobrevibacter thaueri* CW^T (*M. thaueri*), *Methanobrevibacter wolinii* SH^T (*M. wolinii*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*). 200 µl of a full-grown culture were used. The green rectangle marks the same colony in C and D. A: *M. boviskoreani*, B: *M. wolinii*, C: *M. thaueri*, D: *M. thaueri* zoomed in on colony

In all three bottles CFU were formed (see fig. 4.2). *M. thaueri* showed notable less colonies (see table 4.2: 15 for 2432 or 3536), but they were larger than the little dots for the other cultures (fig. 4.2). Another difference between the bottle of *M. wolinii* (fig. 4.2 B) and the bottles of *M. thaueri* (*ibid.* C, D) and *M. boviskoreani* (*ibid.* A) is the agar structure. At *M. wolinii* it is not as smooth and a bit more clumpy. In diskussion auf unterschiedliche technick beim rollen

verweisen -> schlussfolgern wie am besten zu performen ist

However, besides the growing spots inside the bottles, shown in figure 4.2, also the H₂/CO₂ headspace changed. So approximately every two days the gas pressure in bottle dropped from about 1.1 bar to below 0.8 bar and had to be refilled. weist auf anaerobic life hin, ob die spots jetzt wirklich

Verification via 16S-Sequencing, as is done with spread plating, is still pending.

Even though the OD600 of the cultures used was not measured prior to plating, the plating efficiency was determined. This was done on the assumption that the cultures that appeared visually full-grown had an OD600 approximately equal to those listed in tab. 4.1, with the higher value for cell count per mL always being used.

$$\text{Plating efficiency [ppm]} = \frac{\text{counted CFU}}{\text{number of used cells}} \cdot 10^6 \quad (2)$$

The calculation was based on equation 2 and was performed as follows, for example:

$$\text{Plating efficiency} = \frac{2432}{0.2 \text{ ml} \cdot 10.819 \cdot 10^8 \text{ cells} \cdot \text{ml}^{-1}} \cdot 10^6 = 11.24 \text{ ppm}$$

Table 4.2: Counted colony forming unit (CFU) of different *Methanobrevibacter* (*M.*) culture for roll plating technic. 200 µl of a full-grown culture was used for plating. For *M. wolinii* and *M. boviskoreani* was counted a sixteenth of the bottle. Always the higher number of cells per ml from table 4.1 were used as a reference for the cell count in the initial solution.

culture	counted CFU	plating efficiency [ppm]
<i>M. wolinii</i>	2432	11.24
<i>M. thaueri</i>	15	0.13
<i>M. boviskoreani</i>	3536	19.07

4.2 lists the CFU counts per bottle as well as the corresponding plating efficiency. For *M. wolinii* and *M. boviskoreani*, the efficiency is in the lower two-digit ppm range. The efficiency for *M. thaueri* is notable lower, at 0.13 ppm.

4.1.4 Antibiotic sensitivity

Puromycin

M. thaueri and *M. boviskoreani* were cultivated in DSM 1523 media with various concentration of puromycin (Puro): 0 µg/ml, 1 µg/ml, 2 µg/ml, 5 µg/ml, 10 µg/ml and 20 µg/ml. The experiment was performed in triplicates, with each antibiotic (AB) concentration series inoculated with one full-grown pre-culture. The OD600 was taken once a day and plotted logarithmically

against time in figures 4.3 and 4.4. For plotting negative measured OD600 values were set to 0.001. The original data and the adjusted OD600 values are in the appendix (see table B8 and B9). Max die tabellen sind halt extrem lang und gehen dann über drei oder vier seiten, sollen die dann trzdm einfach so in den Anhang? oder kann ich die bei github mit hochladen als csv

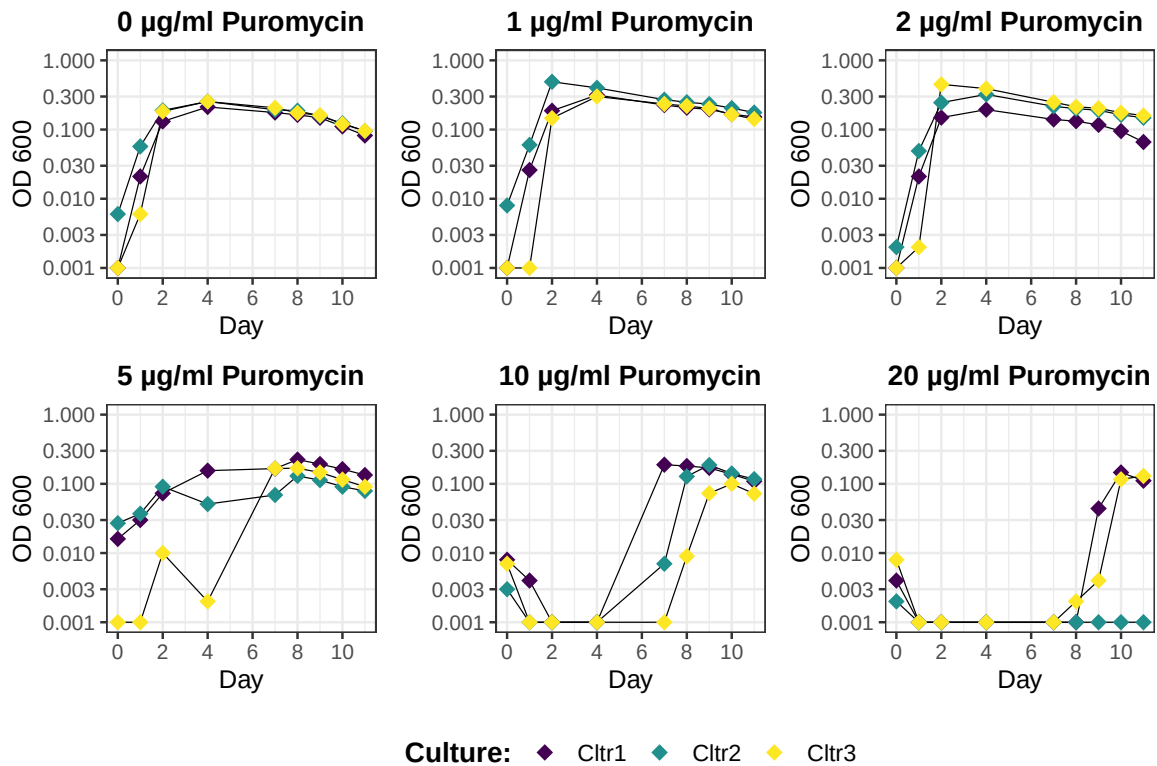


Figure 4.3: Growthcurve in DSM 1523 for *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) for various concentration of puromycin (Puro): 0 µg/ml, 1 µg/ml, 2 µg/ml, 5 µg/ml, 10 µg/ml and 20 µg/ml. Experiment was performed as triplicates (Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

Fig. 4.3 shows that *M. boviskoreani* exhibits a very similar growth pattern for Puro concentrations of 0, 1, and 2 µg/ml. Exponential growth occurs within the first two days. This is followed by a delayed growth phase, such that the growth maximum is reached between days two and four at an OD600 of around 0.3. Until the final measurement day (day 10), the OD decreases continuously to an OD600 value of around 0.1. One difference that can still be noted here, however, is that the variation between the triplicates at 0 µg/ml is smaller than that observed with Puro. In the latter case, culture 3 does not begin exponential growth until one day later as the other triplicates.

At 5 $\mu\text{g/ml}$, the growth curves for Cltr. 1 and Cltr. 2 are generally similar to those just described. However, the two cultures do not grow as quickly. Cltr. 3 shows delayed growth and exhibits greater fluctuations in the OD600 values. Thus, all three triplicates here reach their maximum growth only after 8 days. Nevertheless, the maximum OD600 values here are also slightly lower than in the previous growth curves. Thereafter, the OD600 values decrease continuously.

At 10 and 20 $\mu\text{g/ml}$ Puro concentrations, exponential growth begins with a notable delay. At 5 $\mu\text{g/ml}$, this occurs between 4 and 7 days. The maximum is reached between 7 and 10 days, with maximum OD600 values ranging from approximately 0.1 to 0.25. At 20 $\mu\text{g/ml}$, one replicate shows no growth at all by the end of the measurement (Cltr. 2). The other two replicates do not begin exponential growth until day 8 or 9. The maximum is reached on day 10 at approximately 0.1. By day 11, a slight decrease in the OD600 values can already be observed.

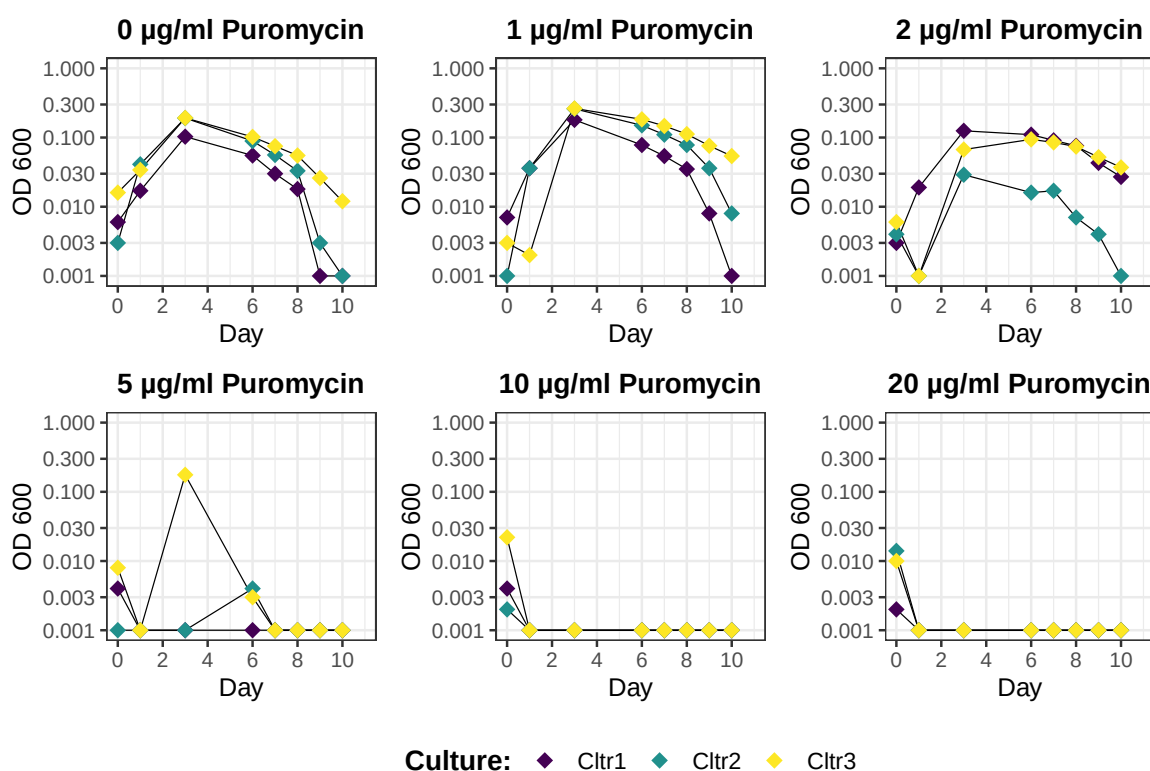


Figure 4.4: Growthcurve in DSM 1523 for *Methanobrevibacter thaueri* CW^T (M. thaueri) for various concentration of puromycin (Puro): 0 $\mu\text{g/ml}$, 1 $\mu\text{g/ml}$, 2 $\mu\text{g/ml}$, 5 $\mu\text{g/ml}$, 10 $\mu\text{g/ml}$ and 20 $\mu\text{g/ml}$. Experiment was performed as triplicates (Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

The growth curves of *M. thaueri* for Puro concentrations of 0 and 1 µg/ml reach their OD600 maximum after 3 days, similar to *M. boviskoreani* (see Figure 4.4). However, the maximum does not exceed 0.3 and is closer to 0.2 than to 0.3.

After that, the OD600 decreases rapidly again, so that at 0 µg/ml, Clt. 1 and 2 already have an OD below 0.003 after 9 days, and Cltr. 3 is also just barely above 0.01 after 10 days. For 1 µg/ml, the final measured OD values range between 0.001 and 0.05.

For 2 µg/ml puromycin, Figure 4.4 shows growth for all three triplicates, reaching its maximum on day 3. However, the increase in OD600 values is significantly lower. Thus, the maximum for Cltr. 2 is below 0.03, and for the others it is around 0.1. This is followed by another decrease in the values, so that the final values range between 0.001 and 0.03.

For the other measured Puro concentrations, in contrast to *M. boviskoreani*, no growth was observed with one exception (Cltr. 3 at 5 µg/ml) (see Fig. 4.3 and 4.4), with the OD600 values remaining below 0.003. For Cltr. 3 at a Puro concentration of 5 µg/ml, an OD600 of more than 0.1 was measured on day 3, but this value dropped back to 0.003 by day 6.

In der Diskussion drauf verweisen, dass man aus der Kurve für thau auch die Lyse gut erkennt, die auch bei den Beobachtungen bei der Kultivierung mit benannt wurden.

Cycloserin

In duplicates it was tested if 200 µg/ml cycloserin (Cyc-Se.) would inhibit the growth of *M. thaueri* in DSM 1523. In all four bottles (two negativ and two with 200 µg/ml Cyc-Se.) visual growth could be detected

4.1.5 Nucleotide-baseanalogica sensitivity test

8-Azahypoxanthin

Similar to the AB sensitivity test, a growth experiment was conducted for the base analogs 8-Azahypoxanthin (8-Aza) for *M. thaueri*, *M. boviskoreani*, *M. millerae*, and 6-Azaauracil (6-Aza) for *M. wolinii*. One difference is that the base analogs do not dissolve in H₂O, so they were dissolved in dimethylsulfoxid (DMSO). Furthermore, different concentrations were selected: 0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml, and one with 1 ml DMSO. The OD600 was measured and plotted on a logarithmic scale against time (see Figures 4.5 through 4.8). The original measured data is in the appendix (Fig. X).

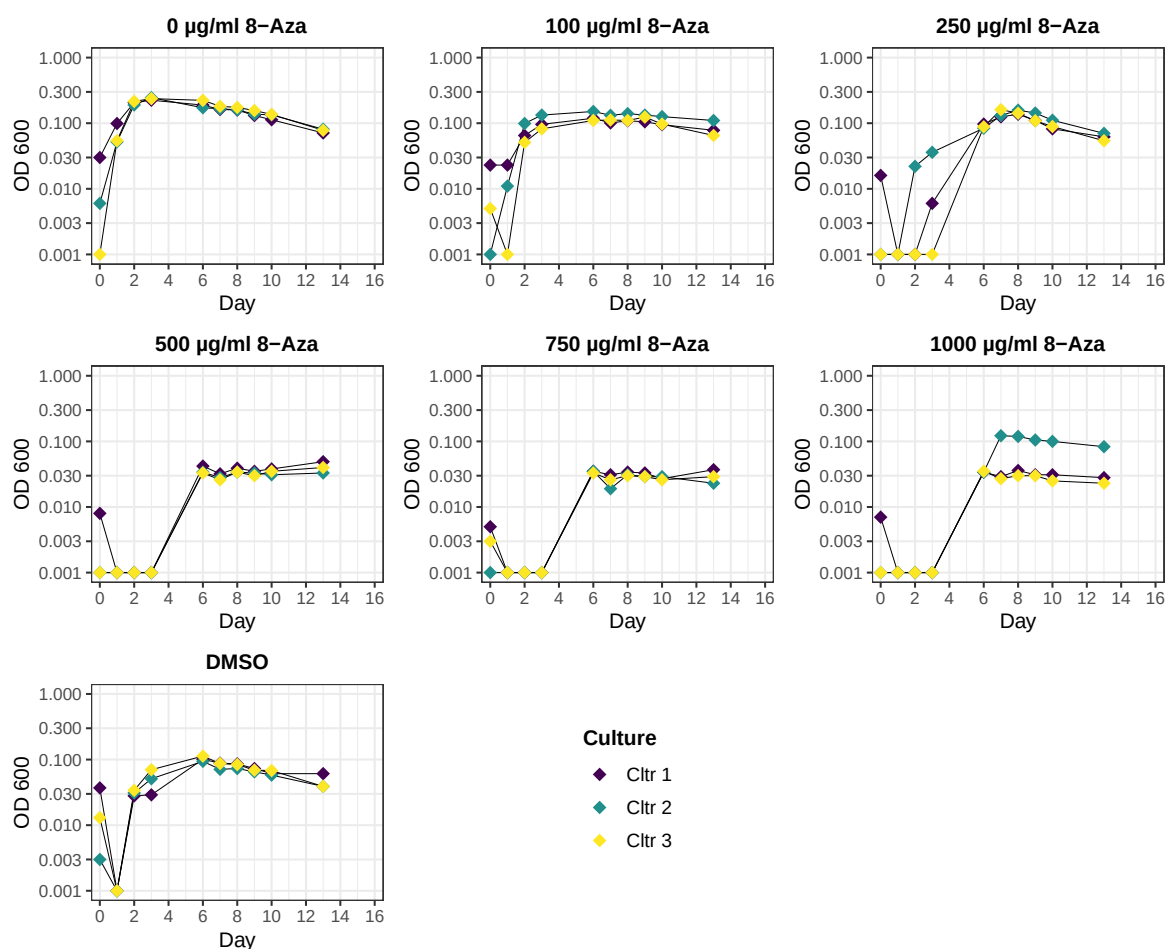


Figure 4.5: Growthcurve in DSM 1523 for *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 8-Aza. Experiment was performed as triplicates(Citr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

4.5 shows a growth curve for the positive control (0 µg/ml 8-Aza) of *M. boviskoreani* that is similar to that of the positive control in Fig. 4.3. Thus, exponential growth occurs within the first two days. On days 3 through 4, the OD600 reaches a maximum of around 0.3, followed by a continuous decline with a final OD600 of just under 0.1 on day 13.

The solvent control (DSMO), in which no base analogs were added but only 1 ml of the solvent DMSO, showed a different growth curve. Initially, none of the triplicates begin to grow immediately; instead, the OD600 values first decrease. On day 3, slight growth begins, reaching a maximum of 0.1 on days 4–6. Afterward, the values decline to approximately 0.03 on day 13.

At the lower concentrations of 8-Aza (100 and 250 µg/ml), *M. boviskoreani* in fig. 4.5 a growth

curve similar to that of the corresponding positive control. However, at 100 µg/ml 8-Aza, the maximum OD600 value is reached within 1 to 3 days, but it is approximately 0.1 instead of just under 0.3. 0.1 is also the maximum OD600 value for the 250 µg/ml concentration; here, growth does not begin until 2 to 4 days later, so that the maximum is not reached until 7 days later.

The three remaining 8-Aza concentration solutions (500, 750, and 1000 µg/ml) exhibit a very similar growth curve. An OD600 below 0.01 was measured for *M. boviskoreani* during the first 4 to 6 days. Then there was an increase to an OD600 of 0.03, which remained constant until the final measurement day (day 13). An exception is Culture 2 in the 1000 µg/ml 8-Aza culture. This culture grows from Day 3 to Day 5 to an OD600 of 0.1, which then decreases slightly until the end of the measurement series.

Figure 4.6 visualizes the growth curves for *M. thaueri*. The first thing that stands out in comparison to the other archaea (fig. 4.5, 4.7, and 4.8) is that the baseline OD600 is higher, at approximately 0.03. With the exception of the positive control (0 µg/ml 8-Aza), no significant increase in the OD600 values was observed. Instead, there was a fairly constant fluctuation around the initial value of 0.03 until the last measurement day (Day 16). An exception here is the triplicate Cltr. 3 in the DMSO sample. Here, an OD600 of 0.3 was measured on Day 2, but it fell back to 0.03 by the next measurement day (Day 4). At the end of the measurement series, this triplicate also ended at approximately 0.07 instead of 0.03 like the other triplicates.

In the positive control, slight growth of *M. thaueri* was detected. However, the growth curve differs quite noticeably from that of the positive control for *M. thaueri* in fig. 4.4. In addition, there is considerable heterogeneity among the individual triplicates.

For example, Cltr. 1 begins to grow slightly from the first day and reaches its maximum OD600 between 0.1 and 0.3 on day 9 (see fig. 4.6). By the end of the measurement, the OD600 decreases continuously to 0.1.

Cltr. 3 begins to grow on Day 8, reaching a maximum OD600 of approximately 0.2 on Day 12. Between Days 13 and 14, the last triplicate then begins to show slight growth, reaching an OD600 of just over 0.1 on Day 16.

The growth curves of *M. millerae* for the positive control (0 µg/ml 8-Aza) and the solvent control (DMSO) have a similar shape for all triplicates, with the measured OD600 being reached on

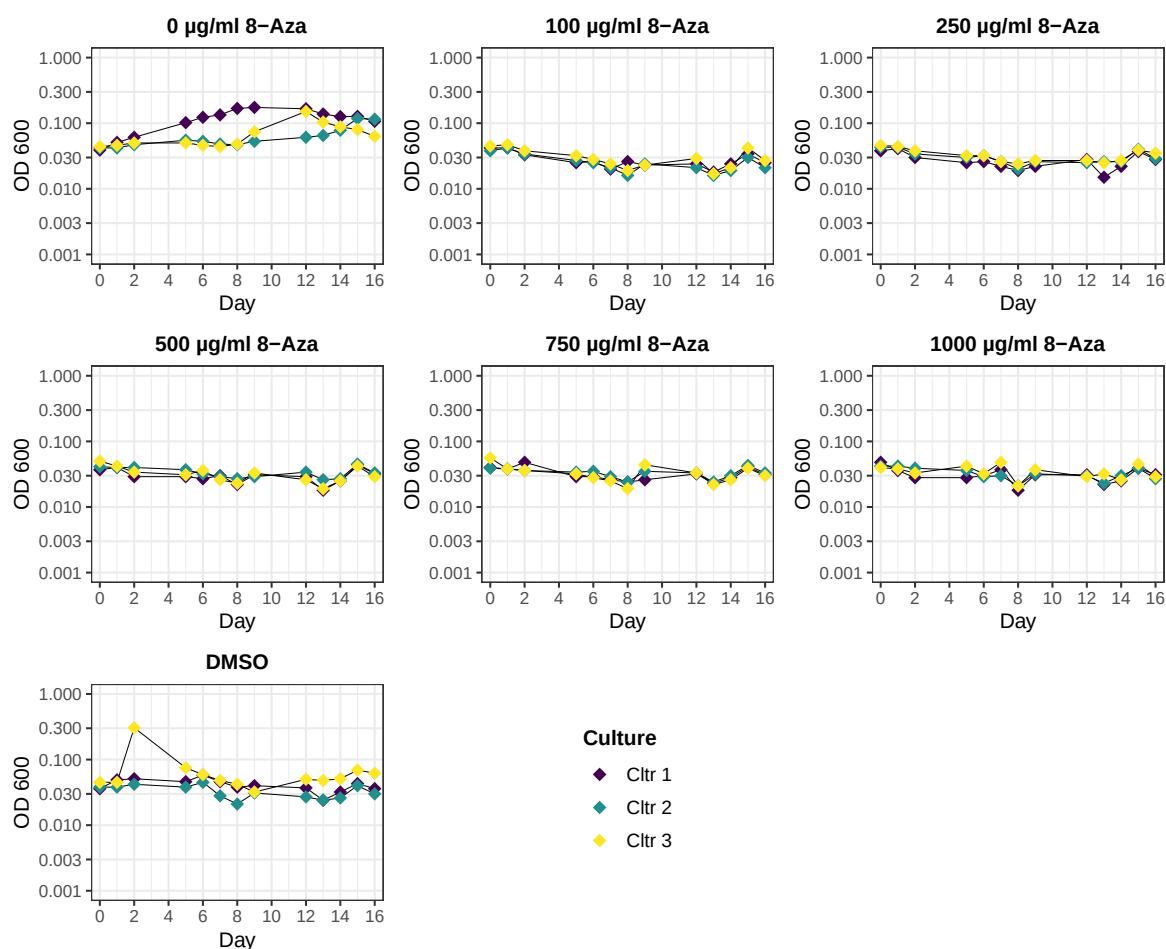


Figure 4.6: Growthcurve in DSM 1523 for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 8-Aza. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

day three, followed by a continuous decline until the end of the measurement series (day 14). However, the curves differ in their OD600 values. The positive control reaches a maximum value of 0.3, while the solvent control reaches just under 0.1. This also results in a significantly lower final OD600 measurement for DMSO (below 0.03) and for the positive control (just under 0.1).

Clear trends for the other samples are generally difficult to discern at first glance; however, most triplicates do not reach an OD600 higher than 0.1.

The 100 µg/ml 8-Aza generally follows the curve shape of the positive and solvent controls, although triplicates Cltr. 1 and 2 reach their maximum OD600 value on Day 4 with a value

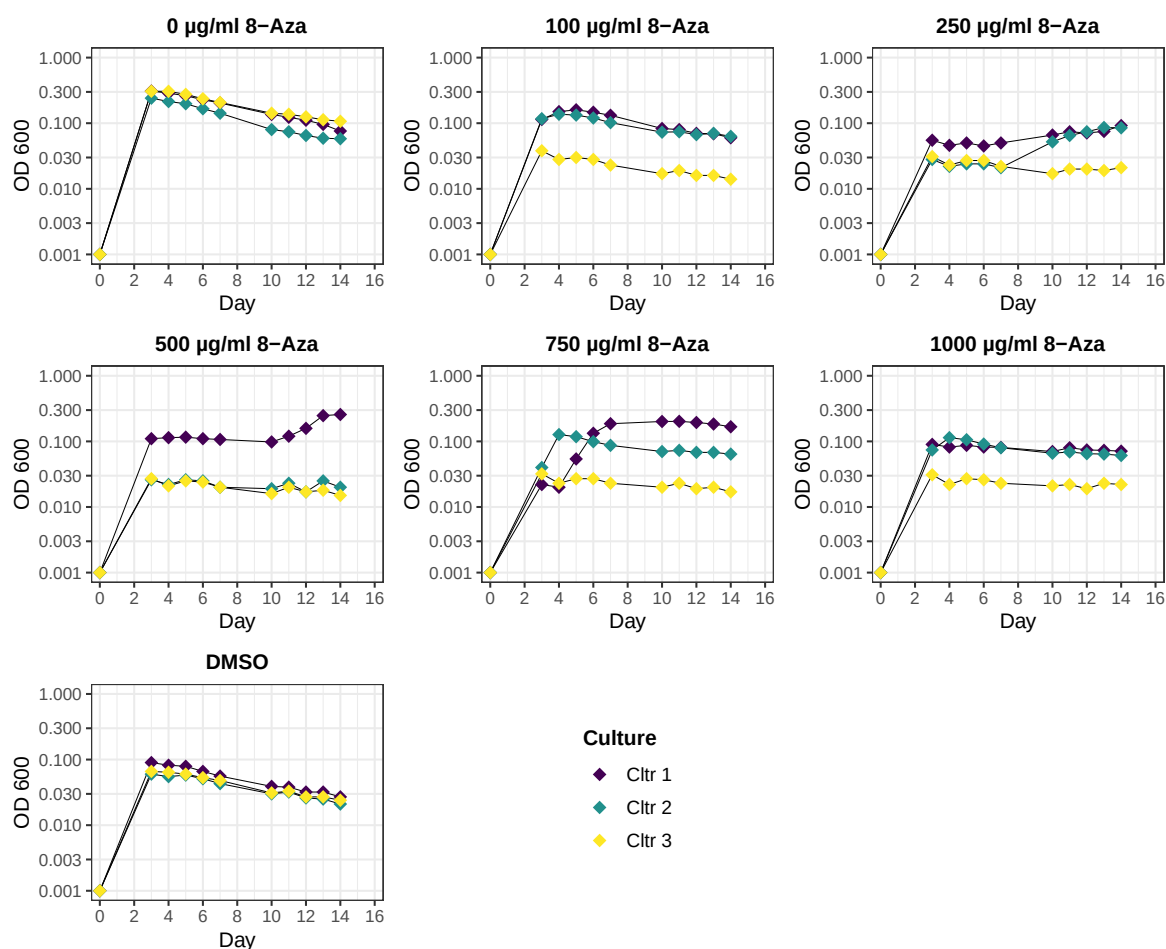


Figure 4.7: Growthcurve in DSM 1523 for *Methanobrevibacter millerae* ZA-10^T (*M. millerae*) for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 8-Aza. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

above 0.1, while Cltr. 3 does not exceed 0.03.

For the 250 µg/ml concentration, the triplicates have a constant OD600 of approximately 0.03, with Cltr. 1 and 2 showing slight growth starting on day 7, reaching approximately 0.1 by the end of the measurement. The triplicates for Ctr. 2 and 3 at the 500 µg/ml concentration show a constant OD600 value of 0.03 starting on Day 3. For Ctr. 1, an OD600 of 0.1 was measured between Days 2 and 10, which increases to just under 0.3 by Day 13.

For the 750 µg/ml and 1000 µg/ml 8-Aza solutions, replicate Ctrl. 3 exhibits a constant value of 0.03 starting on day 2. At 1000 µg/ml, the other two triplicates show a constant OD600 of just under 0.1 starting on day 2, while at 750 µg/ml, only Cltr. 2 shows this value starting on day 2. The Cltr. 1 triplicate in the 750 µg/ml solution shows a value of 0.03 on days 2 and 3, then increases to an OD600 between 0.2 and 0.3 by day 7. This value decreases slightly by

the end of the measurement series.

6-Azaauracil

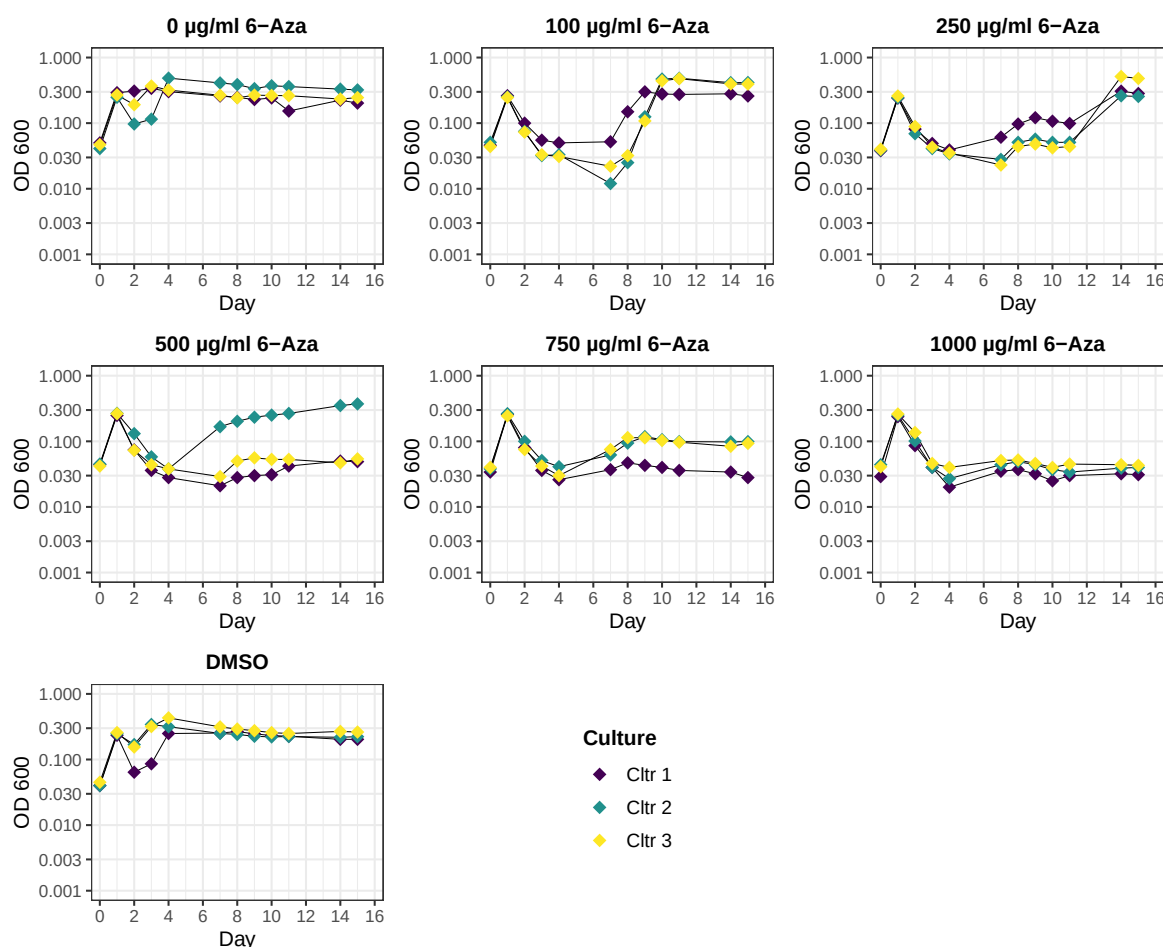


Figure 4.8: Growthcurve in DSM 1523 for *Methanobrevibacter wolinii* SH^T (*M. wolinii*) for various concentration of 6-Azaauracil (6-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 6-Aza. Experiment was performed as triplicates (Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

All growth curves for the 6-Azaauracil (6-Aza) test series by *M. wolinii* (see fig. 4.8) show a jump in the OD600 values within the first day from an initial value of approximately 0.03 to 0.3, followed by a drop that varies in magnitude depending on the sample and results in different curve profiles.

The triplicate Cltr. 1 of the positive control (0 µg/ml) shows no change after Day 1 and maintains a constant OD600 value of around 0.3 until the end of the measurement series. Cltr. 3 dips slightly to approximately 0.2 on Day 2, then remains constant at 0.3 starting on Day 3.

The triplicate Cltr. 2 shows the greatest change after Day 1, remaining at a value of 0.1 on measurement days 2 and 3, before rising to an OD600 of approximately 0.4 on Day 4. The final measured value for this triplicate on Day 15 is approximately 0.3.

The triplicates in the solvent control (DMSO in fig. 4.8) show a more pronounced drop in OD600 values than the positive control. For example, Cltr. exhibits an OD600 of slightly more than 0.1 on Day 2, while Cltr 1 drops to a value below 0.1 on Days 2 and 3. All three triplicates have a fairly stable OD600 of approximately 0.3 starting on Day 4.

In all 6-Aza preparations, a decrease in OD600 values was observed down to a value below approximately 0.03. All triplicates in the 100 µg/ml preparation grew to an OD600 of 0.3 between measurement days 7 and 10, which remained constant until the end of the measurement series.

In the 250 µg/ml preparation, significant growth to an OD600 of more than 0.3 was not observed until between days 11 and 14, even though all triplicates showed a slight increase between days 7 and 8 of approximately 0.03, so that Cltr. 1 had a value of 0.1 and the other two triplicates had values of approximately 0.07.

No further growth was observed for any of the triplicates of the 1000 µg/ml concentration by the end of the measurement series. The same was true for triplicates Cltr. 1 and 3 of the 500 µg/ml 6-Aza concentration and Cltr. 1 at 750 mg/ml.

The OD600 of Cltr 2 in the 500 µg/ml preparation rises from 0.03 on measurement day 4 to approximately 0.2 on measurement day 7, and subsequently to a final OD600 of more than 0.3 by the end of the measurement series.

DMSO

In general all preparation with DMSO in it showed the tendency to be more vulnerable to turn red when taking a sample for photometric measurement. In addition, The decolorization also took longer than in the positive control or other experimental conditions without DMSO, sometimes over an hour.

This interaction of DMSO with the DSM 1523 media was investigated more closely in the *M. thaueri* measurement series. Here, one triplicate with DSM 1523 was prepared with just 1 ml DMSO, but without inoculation (DMSO –, see fig. 4.9).

After the first few days of incubation, a difference in color was observable among the cultures.

Specifically, the DMSO bottles exhibited a red color, in contrast to the yellow color of the other samples in the 8-Azahypoxanthin (8-Aza) measurement series for *M. thaueri*. This is illustrated in fig. 4.9 using selected samples from the triplicate Cltr. 3 after 6 days of incubation. The OD600 was also measured for DMSO, but showed no difference from the baseline OD600 values of the other samples (approximately 0.03 to 0.04).

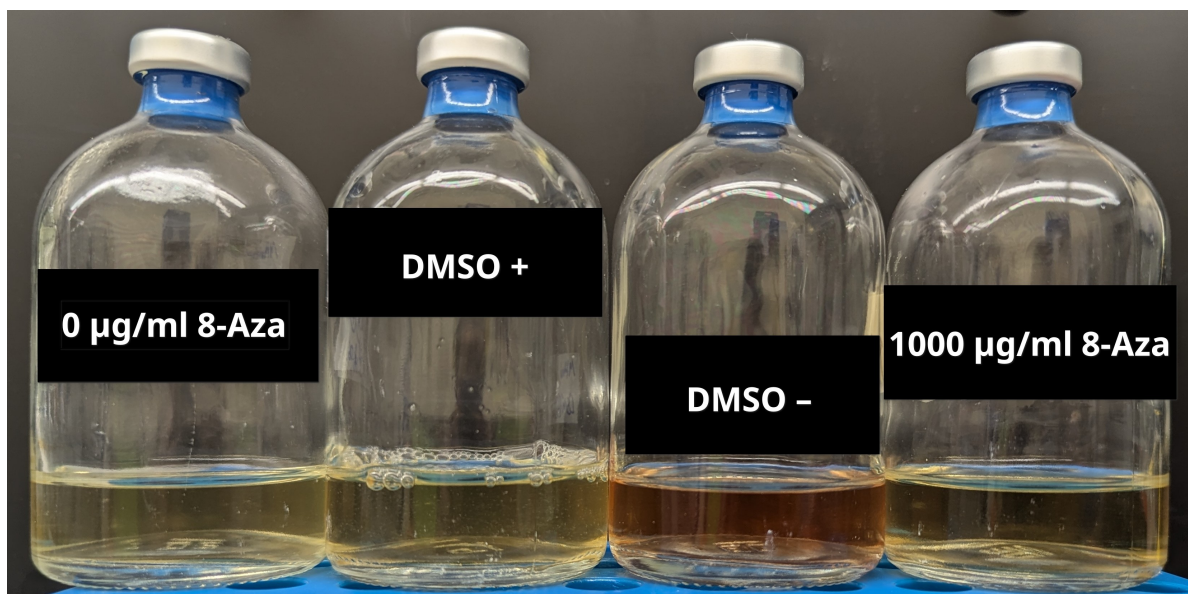


Figure 4.9: Color differences with and without DMSO after 6 days of incubation. Image of various preparations for *M. thaueri* from the same triplicate (Cltr. 3). The positive control (0 µg/ml 8-Azahypoxanthin (8-Aza)) contains only the organism (*M. thaueri*) in DSM 1523, 1000 µg/ml 8-Aza *M. thaueri* and the named base analog, DMSO + *M. thaueri* and 1 ml DMSO; DMSO – also contains 1 ml DMSO but was not inoculated.

4.1.6 Plasmid designing

***pac*-gene**

The *pac*-gene in the initial plasmid (*pac_old*), that should have been taken for the conjugation/ transformation of the archaea, showed a different aminoacids (AA) sequence to the original sequence (*pac_org*) from *Streptomyces alboniger* (see (a) in fig. 4.10). They differ in five AA on the C-terminal end.

These mutation led in to different 3-D structures. In (b) of fig. 4.10 the original *pac* has an antiparallel β -sheet with a big loop. The *pac_old* structure ((c)) shows also an antiparallel β -sheet at this position, but it is longer and has a shorter loop.

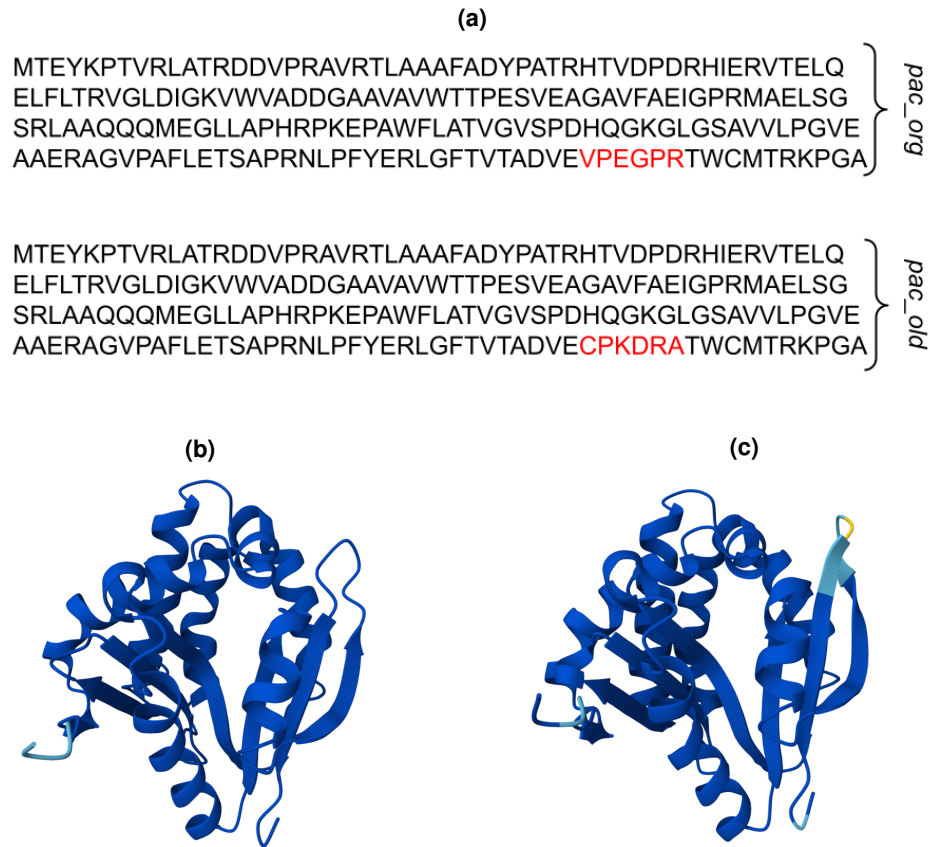


Figure 4.10: aminoacids (AA) sequence of two *pac* genes. (a) shows the AA sequence of the original sequence from *Streptomyces alboniger* (*pac_org*) as well as from the *pac* gene that was located in the plasmid initially planned for conjugation (*pac_old*). The differing section is highlighted in red. (b) The projected 3D structure for *pac_org* calculated by AlphaFold. (c) The projected 3D structure for *pac_old* calculated by AlphaFold.

4.1.7 Transformation/ Konjugation

4.2 Geography

5 DISCUSSION

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6 CONCLUSION

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BIOCHEMICAL TERMS

16S-Sequencing	16S sequencing targets the 16s genes, which code for the 16S ribosomal RNA (rRNA). The 16S rRNA is a part of the prokaryotic ribosome. These 16-s base sequence is specific for the organism.
archaea	Archaea constitute one of the three major domains of life (bacteria, archaea, eukaryotes), and, like bacteria, they belong to the prokaryotes (Fritsche, 2024 , p. 3).
colony	In a microbiological context, a colony is defined as a clearly distinct cluster of cells that can be traced back to a single parent cell through cell division.
complex media	Complex media contain natural, organic molecules, that are not precisely defined chemically (e.g. yeast extract or pepton) So a lot of different microbes can grow on them (Fritsche, 2024 , p. 359).
glovebox	A container that is gas-tight, so that a different atmosphere can be created.
phase-contrast microscope	Phase-contrast microscopy is based on the fact, that objects or medium not only changes the amplitude of light but also the phase of it. The phase-contrast microscope makes this phenomena visual for the human eye (Pawlizak, 2009).

Plating	Cultivate microorganism on a solid medium (contains a gelling agent like agar). Thereby is the plating the process how the organism get distributed on the plate / solid media. Different technics are possible like spreading or pouring (Fritsche, 2024 , p. 364).
pour plating	A plating technique in which the solution containing microorganisms is mixed with the agar-containing culture medium while it is still liquid (at about 45 °C). This mixture is then poured into a Petri dish, where it solidifies(Fritsche, 2024 , p. 364).
ribosome	Ribosomes are a mix out ribosomal RNA (rRNA) and proteins. They play a central role in protein biosynthesis by catalyzing the translation of mRNA into an amino acid sequence.
spread plating	A plating technique in which the solution containing microorganisms is put on the solid agar-containing culture medium. The solution is then spread across the agar plate using a Dragalski spatula (Fritsche, 2024 , p. 364).

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APPENDIX

A Figures

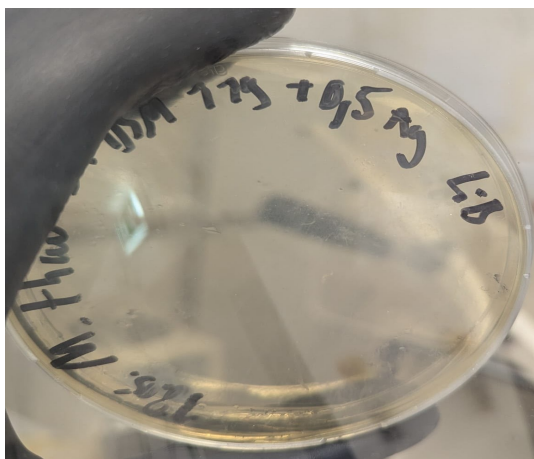


Figure A1: Pour plating result of *Methanobrevibacter thaueri* CW^T (*M. thaueri*) DSM 119 (1% agar). 100 μ l of a full-grown culture have been plated. Cultivated in a jar with H_2/CO_2 gas phase. No growth detected.

In the enteric fermentation process of ruminants CH_4 are formed through methanogenic archaea. One discussed mitigation strategy to reduce methane resulting from cattle farming is to immunize the cattle against these specific archaea ([Malyugina et al., 2025](#)). For developing a specific immunization strategy, the isolated methanogenic archaea strains from cattle rumen has to be well understood. Although some strains have already been isolated from the rumen of cattle, they have hardly been studied to date ([Baca-González et al., 2020](#)). This bachelor's thesis aims to address this issue.

In doing so, the focus will primarily be on two strains: *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*). Besides these two main strains, some experiments were also done on *Methanobrevibacter wolinii* SH^T (*M. wolinii*) and *Methanobrevibacter millerae* ZA-10 T (*M. millerae*). These two methanogenic archaea were also isolated from the rumen of cattle. Goal of the investigation, that took place, was to detect several basic characteristics and try to implement first genetic engineering tools.

B Tables

Table B1: Counted cells for *Methanobrevibacter wolinii* SH^T (*M. wolinii*) with an OD (600 nm) of 0.387. Cells were counted with a phase-contrast microscope and a Buerk counting chamber at 400x magnification. Per chamber filling four square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	138	159	226	233	189
2	260	271	293	265	272.25
3	180	197	216	188	195.25
4	207	191	211	227	209

Table B2: Counted cells for *Methanobrevibacter wolinii* SH^T (*M. wolinii*) with an OD (600 nm) of 0.396. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	177	239	263	254	233.25
2	246	241	267	208	240.5
3	175	176	161	221	183.25
4	217	185	205	180	196.75

Table B3: Counted cells for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) with an OD (600 nm) of 0.235. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:10 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	189	232	250	265	234
2	237	209	224	263	233.25
3	186	253	230	237	226.5
4	146	187	216	262	202.75

Table B8: bovi

culture	day	conc. [μ g/ml]	OD600	OD600 adj.
Cltr. 1	0	0	-0.002	0.001
Cltr. 1	0	1	-0.002	0.001
Cltr. 1	0	2	-0.003	0.001

Cltr. 1	0	5	0.016	0.016
Cltr. 1	0	10	0.008	0.008
Cltr. 1	0	20	0.004	0.004
Cltr. 1	1	0	0.021	0.021
Cltr. 1	1	1	0.026	0.026
Cltr. 1	1	2	0.021	0.021
Cltr. 1	1	5	0.030	0.030
Cltr. 1	1	10	0.004	0.004
Cltr. 1	1	20	-0.008	0.001
Cltr. 1	2	0	0.131	0.131
Cltr. 1	2	1	0.187	0.187
Cltr. 1	2	2	0.149	0.149
Cltr. 1	2	5	0.073	0.073
Cltr. 1	2	10	-0.012	0.001
Cltr. 1	2	20	-0.020	0.001
Cltr. 1	4	0	0.212	0.212
Cltr. 1	4	1	0.315	0.315
Cltr. 1	4	2	0.194	0.194
Cltr. 1	4	5	0.155	0.155
Cltr. 1	4	10	-0.014	0.001
Cltr. 1	4	20	-0.018	0.001
Cltr. 1	7	0	0.176	0.176
Cltr. 1	7	1	0.226	0.226
Cltr. 1	7	2	0.140	0.140
Cltr. 1	7	5	0.166	0.166
Cltr. 1	7	10	0.189	0.189
Cltr. 1	7	20	-0.012	0.001
Cltr. 1	8	0	0.163	0.163
Cltr. 1	8	1	0.207	0.207
Cltr. 1	8	2	0.132	0.132
Cltr. 1	8	5	0.224	0.224
Cltr. 1	8	10	0.181	0.181
Cltr. 1	8	20	-0.004	0.001
Cltr. 1	9	0	0.148	0.148

Cltr. 1	9	1	0.196	0.196
Cltr. 1	9	2	0.117	0.117
Cltr. 1	9	5	0.193	0.193
Cltr. 1	9	10	0.170	0.170
Cltr. 1	9	20	0.044	0.044
Cltr. 1	10	0	0.111	0.111
Cltr. 1	10	1	0.167	0.167
Cltr. 1	10	2	0.095	0.095
Cltr. 1	10	5	0.163	0.163
Cltr. 1	10	10	0.137	0.137
Cltr. 1	10	20	0.146	0.146
Cltr. 1	11	0	0.082	0.082
Cltr. 1	11	1	0.152	0.152
Cltr. 1	11	2	0.066	0.066
Cltr. 1	11	5	0.134	0.134
Cltr. 1	11	10	0.109	0.109
Cltr. 1	11	20	0.111	0.111
Cltr. 2	0	0	0.006	0.006
Cltr. 2	0	1	0.008	0.008
Cltr. 2	0	2	0.002	0.002
Cltr. 2	0	5	0.027	0.027
Cltr. 2	0	10	0.003	0.003
Cltr. 2	0	20	0.002	0.002
Cltr. 2	1	0	0.057	0.057
Cltr. 2	1	1	0.060	0.060
Cltr. 2	1	2	0.049	0.049
Cltr. 2	1	5	0.037	0.037
Cltr. 2	1	10	-0.002	0.001
Cltr. 2	1	20	-0.009	0.001
Cltr. 2	2	0	0.189	0.189
Cltr. 2	2	1	0.490	0.490
Cltr. 2	2	2	0.245	0.245
Cltr. 2	2	5	0.091	0.091
Cltr. 2	2	10	-0.013	0.001

Cltr. 2	2	20	-0.022	0.001
Cltr. 2	4	0	0.254	0.254
Cltr. 2	4	1	0.400	0.400
Cltr. 2	4	2	0.322	0.322
Cltr. 2	4	5	0.051	0.051
Cltr. 2	4	10	-0.020	0.001
Cltr. 2	4	20	-0.024	0.001
Cltr. 2	7	0	0.193	0.193
Cltr. 2	7	1	0.270	0.270
Cltr. 2	7	2	0.217	0.217
Cltr. 2	7	5	0.069	0.069
Cltr. 2	7	10	0.007	0.007
Cltr. 2	7	20	-0.016	0.001
Cltr. 2	8	0	0.187	0.187
Cltr. 2	8	1	0.248	0.248
Cltr. 2	8	2	0.200	0.200
Cltr. 2	8	5	0.130	0.130
Cltr. 2	8	10	0.128	0.128
Cltr. 2	8	20	-0.014	0.001
Cltr. 2	9	0	0.157	0.157
Cltr. 2	9	1	0.233	0.233
Cltr. 2	9	2	0.190	0.190
Cltr. 2	9	5	0.113	0.113
Cltr. 2	9	10	0.187	0.187
Cltr. 2	9	20	-0.017	0.001
Cltr. 2	10	0	0.123	0.123
Cltr. 2	10	1	0.204	0.204
Cltr. 2	10	2	0.163	0.163
Cltr. 2	10	5	0.091	0.091
Cltr. 2	10	10	0.141	0.141
Cltr. 2	10	20	-0.023	0.001
Cltr. 2	11	0	0.095	0.095
Cltr. 2	11	1	0.176	0.176
Cltr. 2	11	2	0.148	0.148

Cltr. 2	11	5	0.079	0.079
Cltr. 2	11	10	0.117	0.117
Cltr. 2	11	20	-0.002	0.001
Cltr. 3	0	0	0.000	0.001
Cltr. 3	0	1	-0.004	0.001
Cltr. 3	0	2	-0.004	0.001
Cltr. 3	0	5	-0.004	0.001
Cltr. 3	0	10	0.007	0.007
Cltr. 3	0	20	0.008	0.008
Cltr. 3	1	0	0.006	0.006
Cltr. 3	1	1	-0.002	0.001
Cltr. 3	1	2	0.002	0.002
Cltr. 3	1	5	-0.008	0.001
Cltr. 3	1	10	-0.010	0.001
Cltr. 3	1	20	-0.004	0.001
Cltr. 3	2	0	0.183	0.183
Cltr. 3	2	1	0.146	0.146
Cltr. 3	2	2	0.450	0.450
Cltr. 3	2	5	0.010	0.010
Cltr. 3	2	10	-0.017	0.001
Cltr. 3	2	20	-0.013	0.001
Cltr. 3	4	0	0.254	0.254
Cltr. 3	4	1	0.300	0.300
Cltr. 3	4	2	0.390	0.390
Cltr. 3	4	5	0.002	0.002
Cltr. 3	4	10	-0.020	0.001
Cltr. 3	4	20	-0.009	0.001
Cltr. 3	7	0	0.206	0.206
Cltr. 3	7	1	0.234	0.234
Cltr. 3	7	2	0.248	0.248
Cltr. 3	7	5	0.167	0.167
Cltr. 3	7	10	-0.006	0.001
Cltr. 3	7	20	-0.002	0.001
Cltr. 3	8	0	0.174	0.174

Cltr. 3	8	1	0.219	0.219
Cltr. 3	8	2	0.213	0.213
Cltr. 3	8	5	0.168	0.168
Cltr. 3	8	10	0.009	0.009
Cltr. 3	8	20	0.002	0.002
Cltr. 3	9	0	0.161	0.161
Cltr. 3	9	1	0.202	0.202
Cltr. 3	9	2	0.203	0.203
Cltr. 3	9	5	0.146	0.146
Cltr. 3	9	10	0.073	0.073
Cltr. 3	9	20	0.004	0.004
Cltr. 3	10	0	0.120	0.120
Cltr. 3	10	1	0.165	0.165
Cltr. 3	10	2	0.175	0.175
Cltr. 3	10	5	0.115	0.115
Cltr. 3	10	10	0.100	0.100
Cltr. 3	10	20	0.116	0.116
Cltr. 3	11	0	0.096	0.096
Cltr. 3	11	1	0.142	0.142
Cltr. 3	11	2	0.159	0.159
Cltr. 3	11	5	0.091	0.091
Cltr. 3	11	10	0.072	0.072
Cltr. 3	11	20	0.130	0.130

Table B9: thau

culture	day	conc. [$\mu\text{g/ml}$]	OD600	OD600 adj.
Cltr. 1	0	0	0.006	0.006
Cltr. 1	0	1	0.007	0.007
Cltr. 1	0	2	0.003	0.003
Cltr. 1	0	5	0.004	0.004
Cltr. 1	0	10	0.004	0.004
Cltr. 1	0	20	0.002	0.002

Cltr. 1	1	0	0.017	0.017
Cltr. 1	1	1	0.036	0.036
Cltr. 1	1	2	0.019	0.019
Cltr. 1	1	5	-0.008	0.001
Cltr. 1	1	10	-0.017	0.001
Cltr. 1	1	20	-0.023	0.001
Cltr. 1	3	0	0.103	0.103
Cltr. 1	3	1	0.180	0.180
Cltr. 1	3	2	0.125	0.125
Cltr. 1	3	5	-0.002	0.001
Cltr. 1	3	10	-0.019	0.001
Cltr. 1	3	20	-0.023	0.001
Cltr. 1	6	0	0.055	0.055
Cltr. 1	6	1	0.078	0.078
Cltr. 1	6	2	0.110	0.110
Cltr. 1	6	5	-0.001	0.001
Cltr. 1	6	10	-0.015	0.001
Cltr. 1	6	20	-0.019	0.001
Cltr. 1	7	0	0.030	0.030
Cltr. 1	7	1	0.054	0.054
Cltr. 1	7	2	0.091	0.091
Cltr. 1	7	5	-0.003	0.001
Cltr. 1	7	10	-0.019	0.001
Cltr. 1	7	20	-0.019	0.001
Cltr. 1	8	0	0.018	0.018
Cltr. 1	8	1	0.035	0.035
Cltr. 1	8	2	0.076	0.076
Cltr. 1	8	5	-0.009	0.001
Cltr. 1	8	10	-0.021	0.001
Cltr. 1	8	20	-0.022	0.001
Cltr. 1	9	0	-0.002	0.001
Cltr. 1	9	1	0.008	0.008
Cltr. 1	9	2	0.043	0.043
Cltr. 1	9	5	-0.024	0.001

Cltr. 1	9	10	-0.035	0.001
Cltr. 1	9	20	-0.035	0.001
Cltr. 1	10	0	-0.013	0.001
Cltr. 1	10	1	-0.009	0.001
Cltr. 1	10	2	0.027	0.027
Cltr. 1	10	5	-0.029	0.001
Cltr. 1	10	10	-0.037	0.001
Cltr. 1	10	20	-0.038	0.001
Cltr. 2	0	0	0.003	0.003
Cltr. 2	0	1	0.001	0.001
Cltr. 2	0	2	0.004	0.004
Cltr. 2	0	5	0.000	0.001
Cltr. 2	0	10	0.002	0.002
Cltr. 2	0	20	0.014	0.014
Cltr. 2	1	0	0.041	0.041
Cltr. 2	1	1	0.036	0.036
Cltr. 2	1	2	-0.004	0.001
Cltr. 2	1	5	-0.005	0.001
Cltr. 2	1	10	-0.013	0.001
Cltr. 2	1	20	-0.003	0.001
Cltr. 2	3	0	0.191	0.191
Cltr. 2	3	1	0.260	0.260
Cltr. 2	3	2	0.029	0.029
Cltr. 2	3	5	0.001	0.001
Cltr. 2	3	10	-0.011	0.001
Cltr. 2	3	20	-0.007	0.001
Cltr. 2	6	0	0.089	0.089
Cltr. 2	6	1	0.150	0.150
Cltr. 2	6	2	0.016	0.016
Cltr. 2	6	5	0.004	0.004
Cltr. 2	6	10	-0.010	0.001
Cltr. 2	6	20	0.000	0.001
Cltr. 2	7	0	0.056	0.056
Cltr. 2	7	1	0.110	0.110

Cltr. 2	7	2	0.017	0.017
Cltr. 2	7	5	0.000	0.001
Cltr. 2	7	10	-0.015	0.001
Cltr. 2	7	20	-0.003	0.001
Cltr. 2	8	0	0.033	0.033
Cltr. 2	8	1	0.078	0.078
Cltr. 2	8	2	0.007	0.007
Cltr. 2	8	5	-0.007	0.001
Cltr. 2	8	10	-0.015	0.001
Cltr. 2	8	20	-0.007	0.001
Cltr. 2	9	0	0.003	0.003
Cltr. 2	9	1	0.036	0.036
Cltr. 2	9	2	0.004	0.004
Cltr. 2	9	5	-0.021	0.001
Cltr. 2	9	10	-0.026	0.001
Cltr. 2	9	20	-0.019	0.001
Cltr. 2	10	0	-0.010	0.001
Cltr. 2	10	1	0.008	0.008
Cltr. 2	10	2	-0.012	0.001
Cltr. 2	10	5	-0.027	0.001
Cltr. 2	10	10	-0.038	0.001
Cltr. 2	10	20	-0.024	0.001
Cltr. 3	0	0	0.016	0.016
Cltr. 3	0	1	0.003	0.003
Cltr. 3	0	2	0.006	0.006
Cltr. 3	0	5	0.008	0.008
Cltr. 3	0	10	0.022	0.022
Cltr. 3	0	20	0.010	0.010
Cltr. 3	1	0	0.034	0.034
Cltr. 3	1	1	0.002	0.002
Cltr. 3	1	2	-0.002	0.001
Cltr. 3	1	5	-0.020	0.001
Cltr. 3	1	10	-0.008	0.001
Cltr. 3	1	20	-0.023	0.001

Cltr. 3	3	0	0.192	0.192
Cltr. 3	3	1	0.263	0.263
Cltr. 3	3	2	0.067	0.067
Cltr. 3	3	5	0.176	0.176
Cltr. 3	3	10	-0.011	0.001
Cltr. 3	3	20	-0.025	0.001
Cltr. 3	6	0	0.102	0.102
Cltr. 3	6	1	0.183	0.183
Cltr. 3	6	2	0.094	0.094
Cltr. 3	6	5	0.003	0.003
Cltr. 3	6	10	0.000	0.001
Cltr. 3	6	20	-0.012	0.001
Cltr. 3	7	0	0.075	0.075
Cltr. 3	7	1	0.147	0.147
Cltr. 3	7	2	0.085	0.085
Cltr. 3	7	5	0.000	0.001
Cltr. 3	7	10	-0.004	0.001
Cltr. 3	7	20	-0.017	0.001
Cltr. 3	8	0	0.055	0.055
Cltr. 3	8	1	0.112	0.112
Cltr. 3	8	2	0.074	0.074
Cltr. 3	8	5	-0.005	0.001
Cltr. 3	8	10	-0.005	0.001
Cltr. 3	8	20	-0.021	0.001
Cltr. 3	9	0	0.026	0.026
Cltr. 3	9	1	0.076	0.076
Cltr. 3	9	2	0.052	0.052
Cltr. 3	9	5	-0.017	0.001
Cltr. 3	9	10	-0.020	0.001
Cltr. 3	9	20	-0.033	0.001
Cltr. 3	10	0	0.012	0.012
Cltr. 3	10	1	0.054	0.054
Cltr. 3	10	2	0.037	0.037
Cltr. 3	10	5	-0.023	0.001

Cltr. 3	10	10	-0.025	0.001
Cltr. 3	10	20	-0.039	0.001

Table B4: Counted cells for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) with an OD (600 nm) of 0.116. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:10 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	255	181	189	171	199
2	127	92	131	193	135.75
3	110	136	153	109	127
4	127	93	109	156	121.25

Table B5: Counted cells for *Methanobrevibacter millerae* ZA-10^T (*M. millerae*) with an OD (600 nm) of 0.378. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	Quadrat_5	$\bar{x}$ cellnumber
1	186	175	173	156	220	182
2	118	204				161
3	211	178	232	211	184	203.2
4	128	128	201	204	160	164.2

Table B6: Counted cells for *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) with an OD(600 nm) of 0.287. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	123	162	166	170	155.25
2	162	228	205	242	209.25
3	170	128	137	213	162
4	187	174	263	237	215.25

Table B7: The total average of counted cells ($\bar{x}$ cellnumber) for each *Methanobrevibacter* (*M.*) culture with corresponding OD at 600 nm (OD600). Counting was done with a phase-contrast microscope and a Buerk or Neubauer improved counting chamber at 400x magnification. $\bar{x}$ cellnumber was calculated as arithmetic mean of shown cellcounts in tab. B1 through B6. The dilution factor takes into account the dilution of the culture being counted.

culture	OD600	Dilution factor	$\bar{x}$ cellnumber
<i>M. wolinii</i>	0.387	20	216.375
<i>M. wolinii</i>	0.396	20	213.438
<i>M. thaueri</i>	0.235	10	224.125
<i>M. thaueri</i>	0.116	10	145.750
<i>M. millerae</i>	0.378	20	177.600
<i>M. boviskoreani</i>	0.287	20	185.438

DECLARATION

I herewith declare that I have composed the present thesis myself and without use of any other than the cited sources and aids. Sentences or parts of sentences quoted literally are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned. The thesis in the same or similar form has not been submitted to any examination body and has not been published. This thesis was not yet, even in part, used in another examination or as a course performance. Furthermore, I declare that the submitted written (bound) copies of the present thesis and the version submitted on a data carrier are consistent with each other in contents.

Place, date

Signature